Lagrange density:

$$\mathcal{L} = \frac{1}{2}\,\mathrm{tr}\left(\dot{\Phi}^2 - (\nabla\Phi)^2 - m^2\Phi^2 - 2g^2\Phi^4\right). \tag{1}$$

Squared commutator:

$$C(t) = \frac{1}{N^4}\sum_{aba'b'}\int d^3\mathbf{x}\;\; \mathrm{tr}\left(\sqrt{\rho}\,[\Phi_{ab}(t,\mathbf{x}),\Phi_{a'b'}]\sqrt{\rho}\,[\Phi_{ab}(t,\mathbf{x}),\Phi_{a'b'}]^\dagger\right). \tag{2}$$

Retarded propagator G_R and Wightman function $\tilde{G}$:

$$\delta_{ab'}\delta_{ba'}G_R(\mathbf{x},t) = \theta(t)\,\mathrm{tr}\left(\rho\,[\Phi_{ab}(\mathbf{x},t),\Phi_{a'b'}]\right) \tag{3}$$

$$\delta_{ab'}\delta_{ba'}\tilde{G}(\mathbf{x},t) = \mathrm{tr}\left(\sqrt{\rho}\,\Phi_{ab}(\mathbf{x},t)\sqrt{\rho}\,\Phi_{a'b'}\right). \tag{4}$$

If Fourier space these take the following form in terms of the spectral function $\rho(k) = \rho(k^0, |\vec{k}|)$:

$$G_R(k) = i\int\frac{d\omega}{2\pi}\frac{\rho(\omega,|\kappa|)}{k^0-\omega+i\epsilon} \tag{5}$$

$$\tilde{G}(k) = \frac{\rho(k)}{2\sinh\frac{\beta k^0}{2}}. \tag{6}$$

For a free field, we have $\rho(\omega,\kappa) = \frac{\pi}{E_\kappa}[\delta(\omega-E_\kappa)-\delta(\omega+E_\kappa)]$, and therefore

$$G_R(k) = \frac{i}{2E_\kappa}\left(\frac{1}{k^0-E_\kappa+i\epsilon} - \frac{1}{k^0+E_\kappa+i\epsilon}\right) \tag{7}$$

$$\tilde{G}(k) = \sum_{s=\pm}\frac{\pi\delta(k^0-sE_\kappa)}{2E_\kappa\sinh\frac{\beta E_\kappa}{2}}. \tag{8}$$